Tilings and Computational Universality

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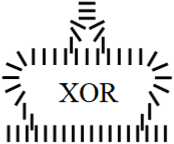

May 12, 2015

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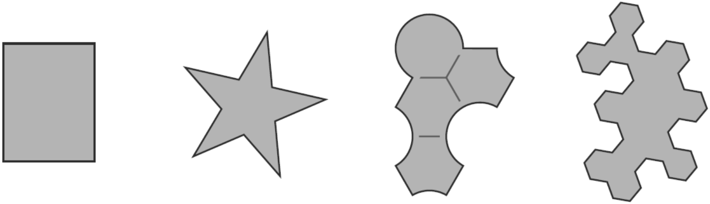

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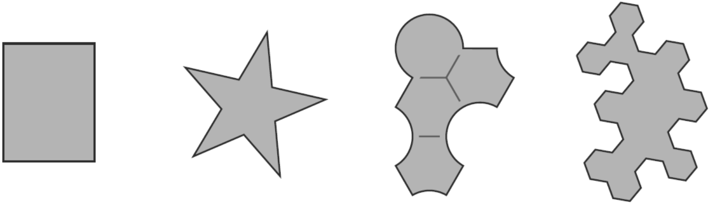

Is there a uniform way to answer this question?

DECISION PROBLEMS

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What if no such algorithm exists? $\rightarrow$ *undecidable*.

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If Wang's conjecture is true, then the Tiling Problem is decidable.

What if the conjecture is false?
The decidability of the Tiling Problem is still unknown.

THE COMPLETION PROBLEM

Problem (Completion Problem)

`Input` **:** *A finite collection of tiles, and an initial, partial tiling of the plane.*

`Output` **:** *Can the initial tiling be completed to a full tiling?*

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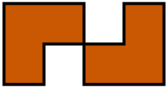
Example

Tile Set	Partial Tiling
	

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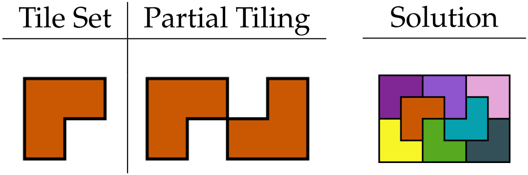

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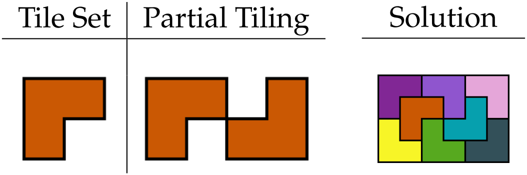

Question (Wang, 1961)

Is the Completion Problem decidable?

THE COLLATZ FUNCTION

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The *Collatz function* is defined as

$$f(n) = \begin{cases} n/2 & \text{if } n \text{ is even} \\ 3n+1 & \text{otherwise.} \end{cases}$$

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For all n, there is an i such that $f^{(i)}(n) = 1$.

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For all n, there is an i such that $f^{(i)}(n) = 1$.

Observation (Erdös)

This is really, really difficult!

Some Collatz Runs

Here are some example runs:

SOME COLLATZ RUNS

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- 5

SOME COLLATZ RUNS

Here are some example runs:

- $5 \to 16$

Some Collatz Runs

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- $5 \to 16 \to 8$

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SOME COLLATZ RUNS

Here are some example runs:

- $5 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$ ($i = 5$)

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For all $n < 5.764 \times 10^{18}$, such an i exists (brute force).
The smallest counterexample to the Polya conjecture is 906,150,257.

COLLATZ DECISION PROBLEM

Problem (Collatz Problem)

`Input`**:** *A natural number, n.*

`Output`**:** *Is there an i such that* $f^{(i)}(n) = 1$*?*

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The Collatz Conjecture, and the decidability of the Collatz Decision Problem are both open.

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Definition

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- At time t, if *any* cell has any other configuration of neighbors, if will be dead at time $t+1$.

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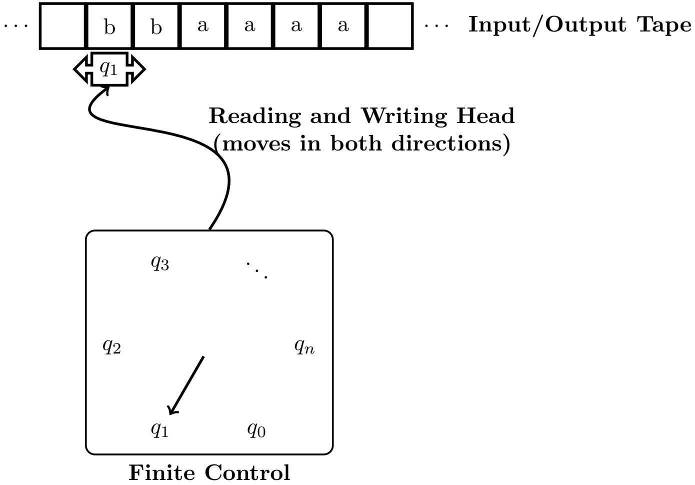
It takes 1103 steps to reach stability!

CONNECTIONS?

What do these problems all have in common?

The Conceptual Turing Machine



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Formally, a *Turing Machine* is a finite deterministic set of transition commands from $Q \times \{0, 1\}$ to $\{0, 1\} \times \{R, L\} \times Q$, where $Q = \{q_0, q_1, \ldots, q_n, H\}$.

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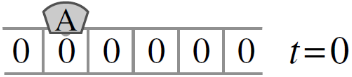
$\leftrightarrow$

$\{(A,0,0,R,B),(A,1,1,R,B),(B,0,1,L,A),$
$(B,1,0,R,C),(C,0,1,R,B),(C,1,0,L,H)\}$

EXAMPLE



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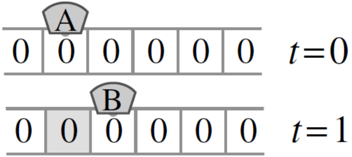

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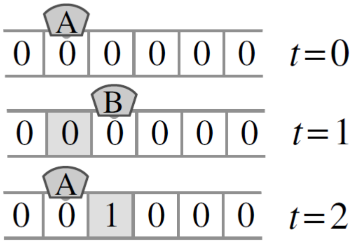

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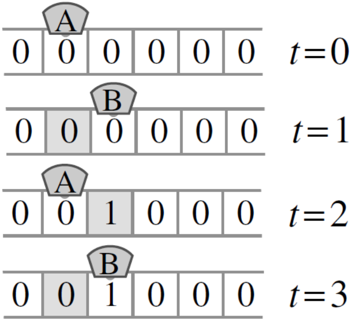

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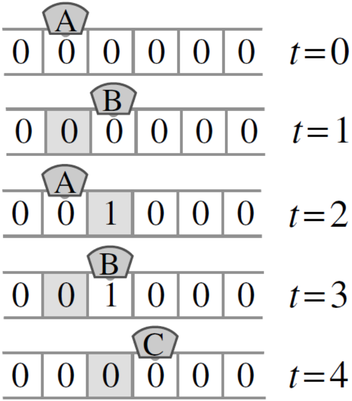

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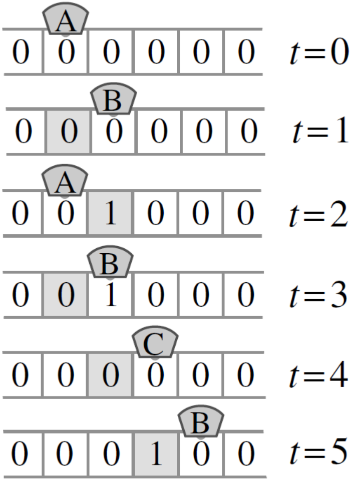
A
0 0 0 0 0 0
t=0
B
0 0 0 0 0 0
t=1
A
0 0 1 0 0 0
t=2
B
0 0 1 0 0 0
t=3
C
0 0 0 0 0 0
t=4
B
0 0 0 1 0 0
t=5

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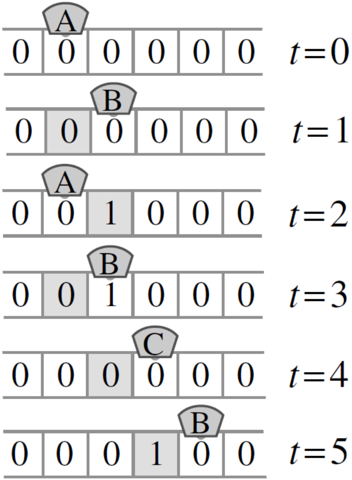

This Turing machine *converges* (enters the Halt state, $\downarrow$) on the empty input tape.

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$$\chi_S(n) = \begin{cases} 1 & \text{if } n \in S \\ 0 & \text{if } n \notin S. \end{cases}$$

CHURCH-TURING THESIS

Observation (Church; Gödel; Kleene; Turing)

All sufficiently powerful models of computation (Partial Recursive Functions, Turing Machines, etc.) all equivalent.

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Definition

A model of computation is said to have *universal computational power* if it can compute those functions computable by Turing Machine.

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Theorem

A set is computable if and only if it is both c.e. and co-c.e.

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- Thus there exists some $i \in \mathbb{N}$ such that $\overline{K} = \{x : \varphi_i(x)\downarrow\}$.
- But then

$$(\forall x)(\varphi_i(x)\downarrow \leftrightarrow x \in \overline{K} \leftrightarrow x \notin K \leftrightarrow \neg\varphi_x(x)\downarrow \leftrightarrow \varphi_x(x)\uparrow).$$

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- Thus there exists some $i \in \mathbb{N}$ such that $\overline{K} = \{x : \varphi_i(x)\downarrow\}$.
- But then

$$(\forall x)(\varphi_i(x)\downarrow \leftrightarrow x \in \overline{K} \leftrightarrow x \notin K \leftrightarrow \neg\varphi_x(x)\downarrow \leftrightarrow \varphi_x(x)\uparrow).$$

- Consider the case where $x = i$.

A NON-COMPUTABLE SET

Theorem

The Halting Set, $K = \{e : \varphi_e(e)\downarrow\}$ is not computable.

Proof.

- Suppose (toward contradiction) that K *is* computable.
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- Consider the case where $x = i$. $\rightarrow$ The world implodes.

□

UPSHOT

Problem (Halting Problem)

`Input`: *A natural number n.*
`Output`: *Is n in K?*

Is this decidable?

COMPLETION REVISITED

Goal: Prove the Completion Problem to be undecidable.

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COMPLETION REVISITED

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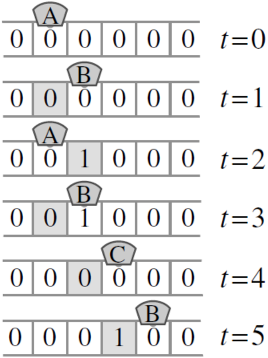
Main Idea: For each Turing Machine, we wish to construct a corresponding set of tiles which will tile the plane if and only if the Turing Machine converges.

Why would this be sufficient?

An example suffices to show the desired encoding.

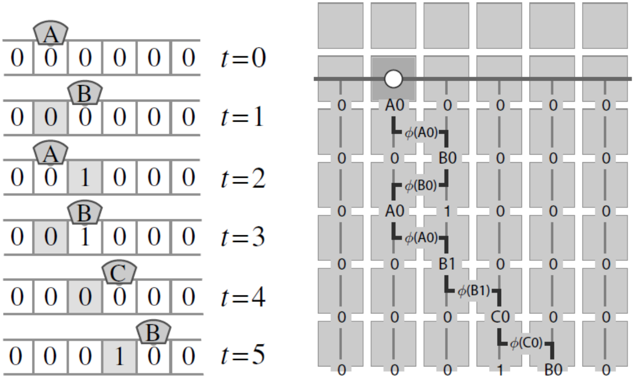
PROOF IDEA

Recall our example Turing Machine from earlier.



Proof Idea

Recall our example Turing Machine from earlier.



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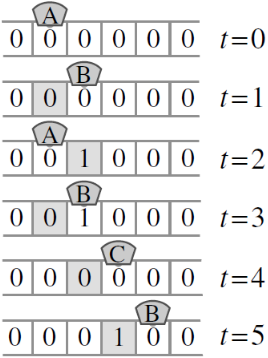
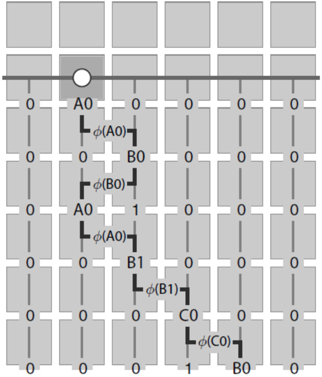
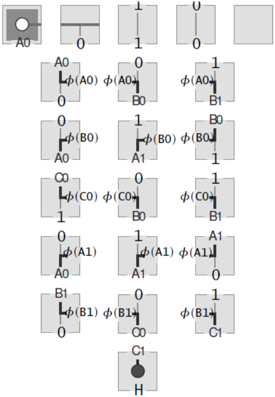




This is our coding scheme!

Proof Idea

Tile Set:



Simulated Machine

PROOF IDEA

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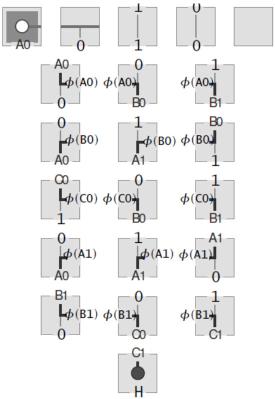

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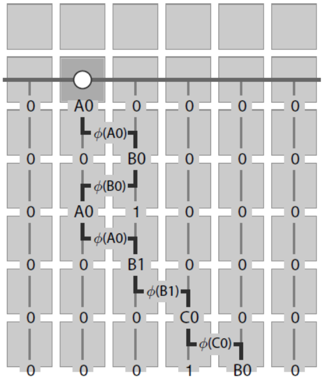

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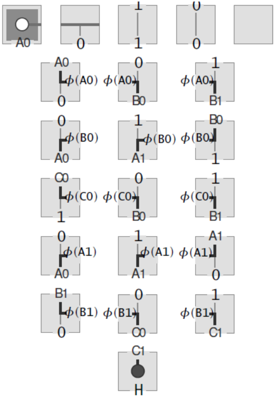

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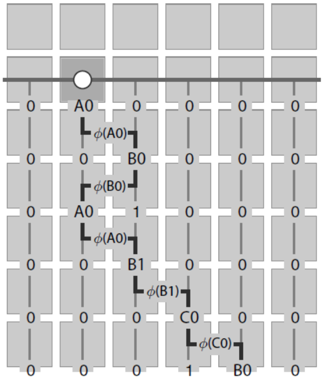

The instances of the Completion Problem have universal computational power, thus proving its undecidability!

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- Let's take a look!
- Turing Machine Life Implementation

Thank you for listening!

REFERENCES

- Robert Berger, *The Undecidability of the Domino Problem*, Thesis (PhD.)-Harvard University, 1965.
- Goodman-Strauss, *Chaim Can't decide? Undecide!*, Notices Amer. Math. Soc. 57 (2010), no. 3, 343356.
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